

Fixed-Wing UAV for Emergency Medical Delivery

AS5213: Conceptual Design of UAVs

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Concept snapshot: time-critical medical delivery, STOL operation, droppable payload, and stable cruise between remote sites.

Placeholder: Isometric CAD render

Presentation Roadmap

Mission and concept

- ▶ Mission profile and requirements
- ▶ Design methodology
- ▶ Aircraft configuration

Verification

- ▶ Performance envelope
- ▶ Stability and eigenvalues
- ▶ Requirements compliance

Technical data

- ▶ Powerplant and energy budget
- ▶ Aerodynamic parameters
- ▶ Mass properties and inertia

Goal

Show that the UAV meets mission range, payload, STOL, and stability targets with a practical conceptual design.

Mission Requirements

Mission objective

A fixed-wing UAV to deliver antivenom and blood products to remote locations with a droppable payload and return capability.

Top-level requirements

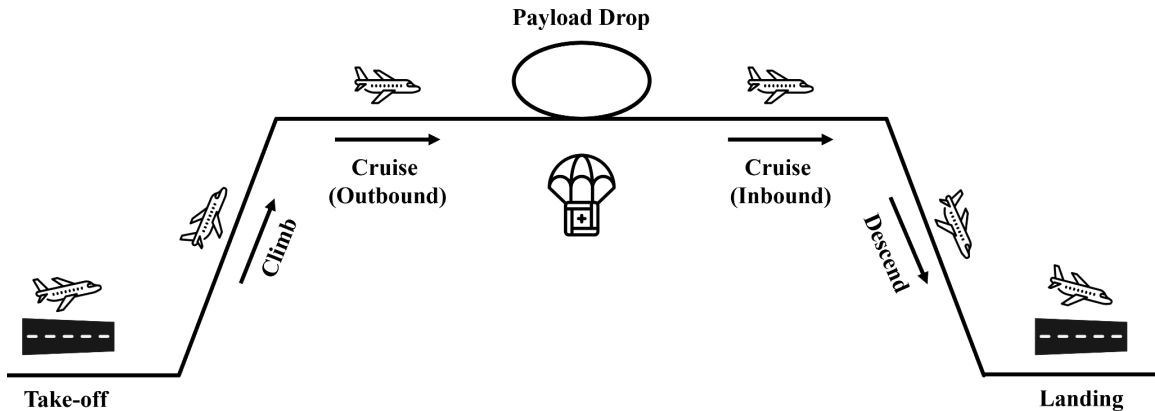
- ▶ Wingspan limited to 2.00 m
- ▶ Payload mass around 1 kg
- ▶ Operational radius 40–50 km
- ▶ Total range 80–100 km
- ▶ Cruise speed 18–25 m/s
- ▶ Takeoff distance below 150 m

Mission phases

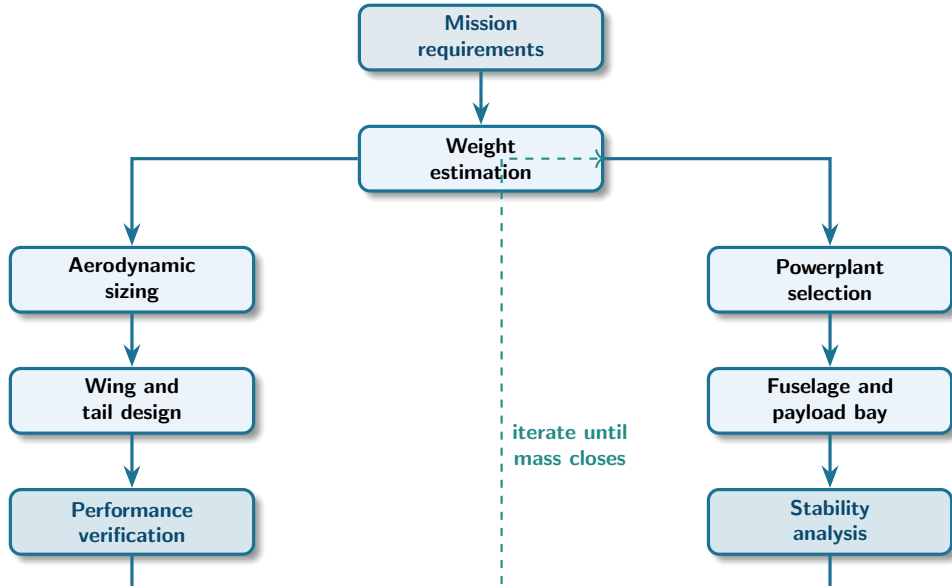
- 1 Takeoff and climb
- 2 Outbound cruise
- 3 Payload drop
- 4 Return cruise
- 5 Descent and landing

Design driver: range and payload have to remain compatible with STOL operation.

Mission Profile



Mission timeline used to size propulsion, energy storage, and payload-drop events.



Technical Data I: Powerplant

Propulsion and energy system

Motor	T-Motor AM480 900KV
Configuration	Dual tractor motors
Maximum power	450 W at 80% throttle
Supply voltage	11.1 V (3S LiPo)
Peak current	26 A
ESC	Hobbywing Skywalker 50A V2
Propeller	APC 12×6E

Power budget

Climb power: 361.46 W	Avionics: 6.61 W
Battery efficiency: 85%	Motor efficiency: 82%

Component imagery

**Motor / propeller / battery
images were not provided**

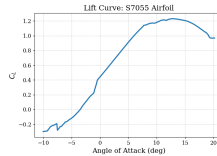
Installation note

Dual-tractor wing-mounted propulsion with the battery placed to help CG control.

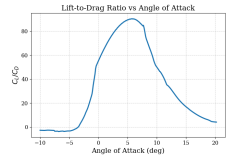
Technical Data II: Aerodynamics

Wing and drag data

Airfoil	Selig S7055
Planform	Rectangular, high-wing
Aspect ratio	7.53
Wing area	0.5307 m ²
Wingspan	2.00 m
Mean chord	0.2653 m
Zero-lift drag	$C_{D_0} = 0.02873$
Oswald efficiency	$e = 0.85$
Max L/D	12.55



Lift curve



C_L/C_D vs. angle of attack

Key result

$C_{L_\alpha} = 5.73/\text{rad}$, cruise $C_L \approx 0.65\text{--}0.75$, and estimated stall speed 12.4 m/s.

Technical Data III: Mass Properties

Mass breakdown

MTOW	6.59 kg
Empty weight	3.29 kg
Battery	0.987 kg
Payload	1.20 kg
Avionics	0.90 kg
Empty fraction	0.50
Static margin	$\approx 14\%$ MAC

Geometry

Fuselage length	1.10 m	
Fuselage diameter	0.15 m	
Tail boom length	0.70 m	
CG location	264 mm	from

Empennage and controls

Horizontal tail area	0.133 m ²
Horizontal tail span	0.773 m
Tail arm	0.70 m
Tail airfoil	NACA 0012
Tail setting angle	-0.31°
Vertical tail area	0.080 m ²

Design note

The payload bay is placed near the CG and supports droppable payload release without destabilizing the aircraft.

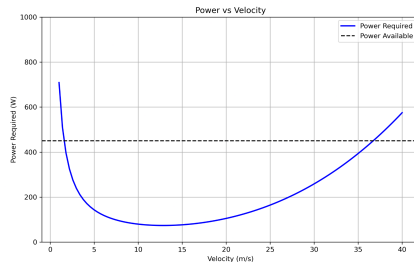
Performance I: Power Required

Performance summary

- ▶ Cruise speed: 18–22 m/s
- ▶ Maximum range: 83.15 km
- ▶ Endurance: 2 h 5 min
- ▶ Service ceiling: 2000 m
- ▶ Maximum speed: 28 m/s

Power margin

The required power remains below available power across the practical cruise band.



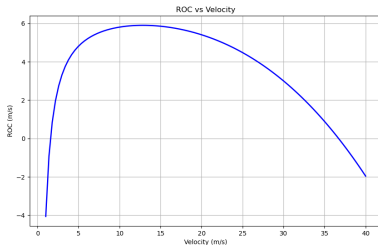
Performance II: Rate of Climb

Climb performance

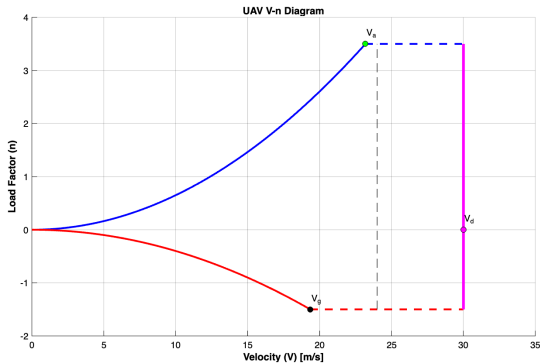
- ▶ Rate of climb: 3.5 m/s
- ▶ Optimal climb speed: 16 m/s
- ▶ Climb angle: 15–20°
- ▶ Takeoff distance: < 150 m

Takeaway

Climb performance is adequate for the mission and remains inside the available power margin.



Performance III: Load Factor and Takeoff Envelope



Positive maneuver envelope before payload release.

Takeoff and landing

Takeoff distance	< 150 m
Liftoff speed	13.6 m/s
Landing distance	< 120 m
Approach speed	15 m/s
Positive limit load	3.5
Negative limit load	-1.5

Operational takeaway

The aircraft satisfies the STOL requirement while retaining a practical speed margin for climb, cruise, and landing.

Stability I: Static Derivatives

Longitudinal stability

C_{m_α}	-0.487 rad^{-1}
C_{m_0}	0.0125
C_{m_q}	-14.94 rad^{-1}
C_{L_q}	$+5.66 \text{ rad}^{-1}$
C_{D_α}	$+0.104 \text{ rad}^{-1}$

Interpretation

Negative C_{m_α} indicates static longitudinal stability, and the CG remains ahead of the neutral point.

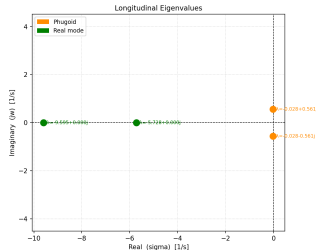
Lateral-directional stability

C_{n_β}	$+0.0838 \text{ rad}^{-1}$
C_{l_β}	-0.0111 rad^{-1}
C_{l_p}	-0.659 rad^{-1}
C_{n_r}	negative, stabilizing
Roll damping	strong

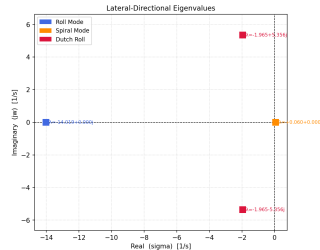
Interpretation

The sign pattern is consistent with weathercock stability, roll damping, and a conventional tail layout.

Stability II: Eigenvalue Analysis



Longitudinal modes: phugoid and real mode.



Lateral-directional modes: roll, spiral, and Dutch roll.

Mode	Key result	Interpretation
Phugoid	lightly damped	expected for small UAV
Roll mode	fast decay	strong roll stabilization
Dutch roll	stable complex pair	acceptable damping
Spiral mode	slightly unstable / slow	monitor in trim and control tuning

Isometric CAD Views

Isometric view placeholder

Side view placeholder

Top view placeholder

Front view placeholder

Integration notes

The final CAD package should show the payload bay, propulsion placement, landing gear clearance, control surfaces, and avionics access.

Requirements Compliance Summary

Requirement	Target	Achieved	Status
Maximum wingspan	2.00 m	2.00 m	✓
Payload capacity	1–2 kg	1.20 kg	✓
Operational radius	40–50 km	41.6 km	✓
Total range	80–100 km	83.15 km	✓
Endurance	1–2 h	2 h 5 min	✓
Cruise speed	18–25 m/s	18–22 m/s	✓
Takeoff distance	< 150 m	< 150 m	✓
Static stability	positive margin	≈ 14% MAC	✓

Overall conclusion: the conceptual UAV meets the mission, range, payload, STOL, and stability goals.

Thank you

Questions?